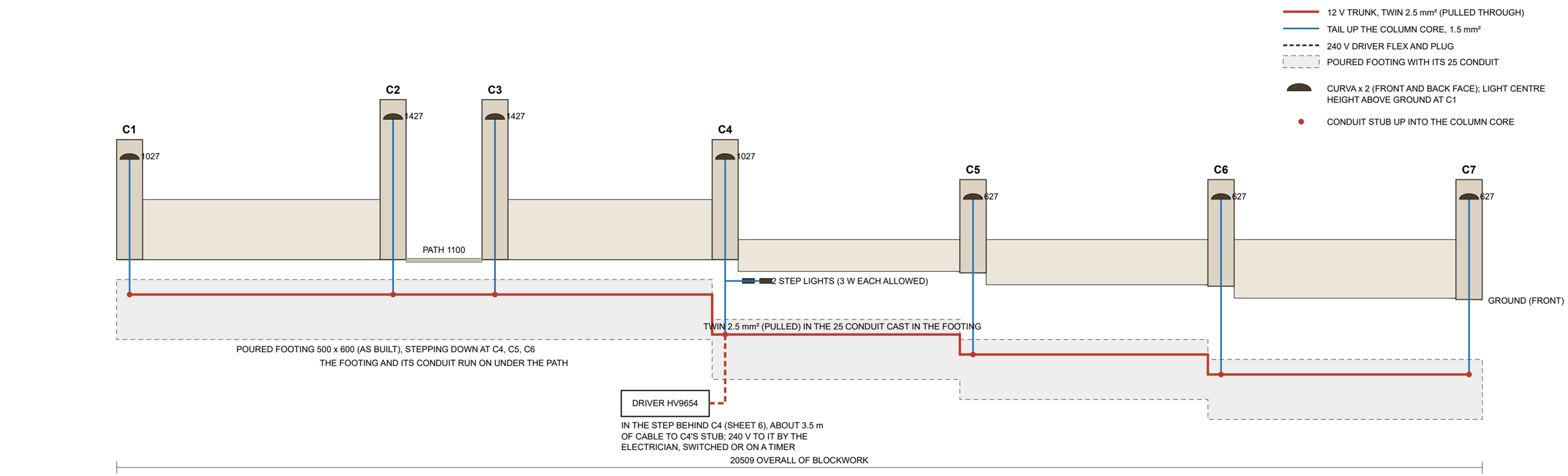
BLOCK WALL BW01 - COLUMN LIGHTS, HAVIT CURVA HV3251T-AB-12V: WIRING AND MOUNTING SCHEMATIC

PRELIMINARY - FOR THE ELECTRICIAN TO CONFIRM

13 NOV 2026 · set-out as built: opening 1140 between the C2 and C3 block faces; walls tiled 20 off the blocks; 25 conduit cast in the footings and twin 2.5 mm² pulled through, as built; driver formed into the step behind C4 · schematic, not to scale except where stated

A. WIRING LAYOUT - WALL SEEN FROM THE FRONT (C1 LEFT). LENGTHS ALONG THE WALL TO 1:60, HEIGHTS DRAWN 1.5 TIMES AS TALL.



VOLTAGE AT THE LIGHTS ON THE 2.5 mm² AS PULLED (HAVIT: 11.40 V OR MORE, NO MORE THAN 25 m OF CABLE FROM THE DRIVER)

Column	Light centre (mm)	Driver in the step behind C4, step lights on (volts / cable)	Same, step lights off	For reference: driver at the GPO 18 m south of C1 (volts / cable)
C1	1027	11.65 V / 15 m	11.68 V	**10.92 V / 21 m**
C2	1427	11.68 V / 11 m	11.70 V	**10.73 V / 25 m**
C3	1427	11.70 V / 10 m	11.72 V	**10.66 V / 27 m**
C4	1027	11.78 V / 6 m	11.80 V	**10.53 V / 30 m**
C5	627	11.70 V / 9 m	11.72 V	**10.45 V / 34 m**
C6	627	11.64 V / 14 m	11.67 V	**10.39 V / 38 m**
C7	627	11.62 V / 17 m	11.64 V	**10.36 V / 42 m**

With the driver in the step every light gets 11.62 V or more and no light is more than 17 m of cable from it - inside Havit's limits. With the driver at the GPO 18 m south of C1 (right-hand column) every light failed; a bigger driver would not have helped, because it still puts out 12 V.

THE WORKING (DRIVER IN THE STEP, STEP LIGHTS ON)

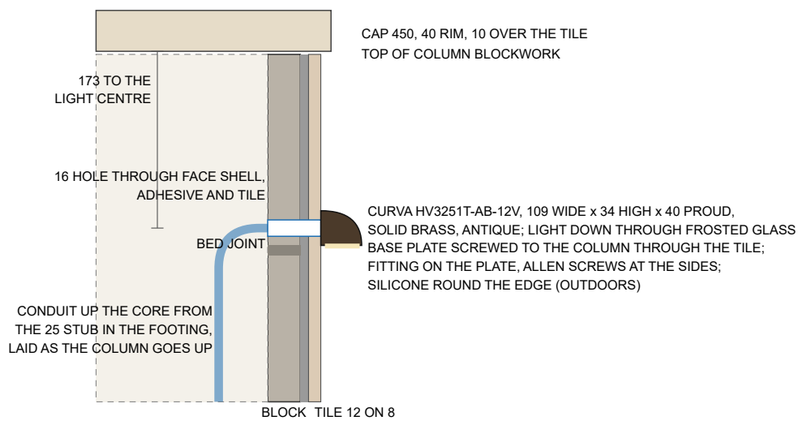
Trunk run	Length	Current
Driver in the step to C4's stub	3.50 m	4.00 A
C4 to C3	3.86 m	1.50 A
C3 to C2, on under the path	1.53 m	1.00 A
C2 to C1	3.96 m	0.50 A
C4 to C5	3.92 m	1.50 A
C5 to C6	3.92 m	1.00 A
C6 to C7	3.73 m	0.50 A

Each Curva 3 W = 0.25 A at 12 V; a tail feeds 2 lights. Drop out and back = 2 × 0.0175 × length × current ÷ 2.5 mm², added run by run; tails 1.5 mm². Lengths follow the footing's steps; the conduit's real depth may add a little. The driver's size does not change these figures: a 12 V constant voltage driver puts out 12 V whatever its rating; the drop comes from the current and the cable.

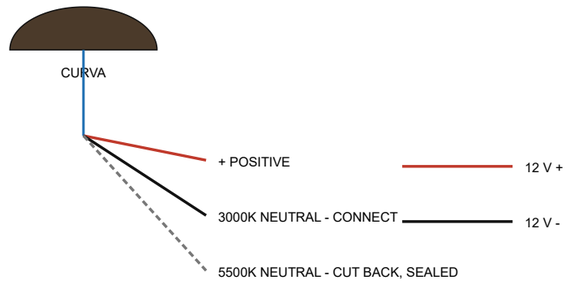
NOTES

1. 14 × Havit Curva HV3251T-AB-12V (solid brass, antique brass), 3 W, 240 lm at 3000K, IP65, 12 V DC, on both faces of each column, bottom of each light on the bed joint under the top block (light centre 173 below the top of the column blockwork); see the drill pattern, sheet BW01-5. Wired fixed at warm white 3000K.
2. Driver: 1 × Havit HV9654, 100 W weatherproof (IP66), 12 V DC constant voltage, flex and plug, formed into the end of the step behind C4 in an open, drained niche with a weatherproof outlet (sheet 6); 240 V to it by the electrician. Load: 14 × 3 W + 2 step lights allowed at 3 W = 48 W; Havit sizes drivers for light fittings at 50% of their rating, so 100 W (their recommended maximum load 80 W); a 150 W driver also suits but does not raise the voltage at the lights. Switched or on a timer on the 240 V side; not dimmed.
3. As built: 25 conduit cast in the footings, up into every column core (C1 to C7), with twin 2.5 mm² pulled through, west to C1 and east to C7. The driver feeds it at C4: a twin 2.5 mm² in 25 conduit from the step to C4's core, joined in a reachable draw box; the run from C1 back to the GPO is no longer used for 12 V. The footing runs on under the path, so the cable crosses the C2 to C3 opening inside it.
4. From each stub, carry the conduit up the column core to both light positions as the column goes up, before core fill, clear of the N12 bars - certifier to accept. 1.5 mm² tails in it to the lights.
5. Make every joint where it can be reached again (behind a fitting or in a draw box), never buried in core fill or the footing. Havit joints: solder, then adhesive-lined heat shrink or a gel cap, or Havit IP67 connectors.
6. Havit: installed by a licensed electrician to AS/NZS 3000; mains off before work; gloves when handling brass fittings; silicone round each fitting outdoors. Havit lists aluminium, 304 stainless, powder-coated brass and 316 stainless fittings as not for use within 5 km of salt water, pools included; the antique brass Curva is solid brass - confirm with Havit for this site.
7. The 2 step lights on the step riser by C4: fitting not chosen; on this circuit they cost about 0.2 V at the far lights.

B. MOUNTING DETAIL - SECTION THROUGH A COLUMN FACE AT A LIGHT, 1:6



C. WIRING AT EACH LIGHT (HAVIT), FIXED AT WARM WHITE 3000K



ALL 16 LIGHTS IN PARALLEL ON THE ONE 12 V DC DRIVER. NEVER CONNECT A LIGHT TO 240 V.
JOINTS SOLDERED, COVERED WITH ADHESIVE-LINED HEAT SHRINK OR A GEL CAP,
OR HAVIT IP67 CONNECTORS HV9911 / HV9912 / HV9913 (HAVIT).
4000K WOULD JOIN BOTH NEUTRALS; NOT USED HERE.